

- Document Control
- System Name: Application
- Version:
- Prepared By:
- Date:
- Approved By:
- System Overview
 - The Application processes source CSV data, loads it into Delta tables (customer, order), applies transformation and cleansing, creates summary and aggregate spend tables, and manages SCD type 2 history using Delta Live Tables.
- User Roles
 - Data Engineer
 - Data Analyst
 - Administrator
- Functional Scope
 - Ingest customer and order data from CSV files.
 - Clean and transform ingested data.
 - Manage Slowly Changing Dimension (SCD) type 2 ordersummary table.
 - Aggregate and store customer spending.
 - Automate ETL pipeline using Delta Live Tables.
- Functional Requirements

FR-APP-001 — Ingest Customer and Order Data

- Description: Read source CSV data from defined volumes and load into 'customer' and 'order' Delta tables.
- Preconditions: Source CSV files are available at specified paths.
- Main Flow:
 1. Read customer CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
 2. Read order CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
 3. Load data into their respective Delta tables.
- Alternate Flows:
- Acceptance Criteria: Data successfully loaded into 'customer' and 'order' tables.

FR-APP-002 — Schema Definition for Ingested Tables

- Description: Enforce schema for customer and order tables.
- Preconditions: Tables created or already exist.
- Main Flow:

1. Define schema for customer: CustId, Name, EmailId, Region.

2. Define schema for order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

- Alternate Flows:
- Acceptance Criteria: Schema matches as defined and is enforced at ingestion.

FR-APP-003 — Create TotalAmount Column

- Description: Add a 'TotalAmount' column to 'order' by multiplying PricePerUnit and Qty.
- Preconditions: Data loaded in 'order' table.
- Main Flow:

1. Compute $\text{TotalAmount} = \text{PricePerUnit} * \text{Qty}$.

2. Add 'TotalAmount' column to 'order'.

- Alternate Flows:
- Acceptance Criteria: TotalAmount is present and populated correctly.

FR-APP-004 — Data Cleansing

- Description: Remove null and duplicate records from 'customer' and 'order' tables.
- Preconditions: Data loaded in Delta tables.
- Main Flow:

1. Identify records with null values and remove them.

2. Detect and remove duplicate records.

- Alternate Flows:
- Acceptance Criteria: No null or duplicate records remain.

FR-APP-005 — Create Ordersummary Table

- Description: Create 'ordersummary' table in catalog='gen_ai_poc_databrickscoe', schema='sdlc_wizard' if it does not exist.
- Preconditions: Delta catalog and schema exist.
- Main Flow:

1. Check if 'ordersummary' exists; create if not.

2. Ensure schema: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.

- Alternate Flows:
- Acceptance Criteria: Ordersummary table exists with specified schema.

FR-APP-006 — Load Ordersummary with SCD Type 2

- Description: Join customer and order data on CustId and load into SCD type 2 table 'ordersummary'.
- Preconditions: Both customer and order tables populated.
- Main Flow:

1. Perform join on CustId.

2. Load resultant data into 'ordersummary' as SCD type 2.

- Alternate Flows:
- Acceptance Criteria: Ordersummary table reflects SCD2 changes and maintains history.

FR-APP-007 — SCD Type 2 Update Logic

- Description: Update 'ordersummary' as SCD type 2 upon changes in customer table (set old records inactive, add new active records, update StartDate/EndDate).
- Preconditions: Ordersummary table exists and is populated.
- Main Flow:

1. Detect changes in customer table.

2. Mark old 'ordersummary' records inactive, add new active records.

3. Update StartDate and EndDate fields to maintain history.

- Alternate Flows:
- Acceptance Criteria: SCD2 logic works correctly and history is retained.

FR-APP-008 — Create CustomerAggregateSpend Table

- Description: Create 'customeraggregatespend' table with columns Name, TotalAmount, Date in specified catalog/schema if not exists.
- Preconditions: Catalog and schema exist.
- Main Flow:

1. Check/create 'customeraggregatespend'.

2. Ensure columns: Name, TotalAmount, Date.

- Alternate Flows:
- Acceptance Criteria: Table exists with correct structure.

FR-APP-009 — Aggregate and Load Customer Spending

- Description: Aggregate TotalAmount by Name and Date from 'ordersummary' and load into 'customeraggregatespend'.
- Preconditions: Ordersummary table contains data.
- Main Flow:

1. Perform aggregation: group by Name and Date, sum TotalAmount.
2. Load aggregated results into 'customeraggregatespend'.

- Alternate Flows:
- Acceptance Criteria: Aggregated data loaded and correct.

FR-APP-010 — Implement with Delta Live Tables

- Description: Use Delta Live Tables framework to implement steps 1–10.
- Preconditions: Delta Live Tables runtime available.
- Main Flow:

1. Build and orchestrate all functional pipeline steps using Delta Live Tables.

- Alternate Flows:
- Acceptance Criteria: Entire pipeline automated and runs as per requirements.
- Business Rules
 - Enforce specified schema for each table.
 - Remove records with nulls and duplicates before further processing.
 - Update SCD type 2 table according to customer changes.
 - Aggregate on Name and Date fields only.
- Error Handling
 - Log ingestion errors for missing files or corrupted data.
 - Reject records with missing mandatory fields.
 - Log aggregation failures.
- Dependencies
 - Availability and accessibility of CSV source files at specified paths.
 - Working Delta Lake environment.
 - Delta Live Tables framework.

- Acceptance Criteria
- All functional requirements pass user and system testing.
- SCD type 2 history is accurately maintained.
- Aggregation and summary tables are correct and up-to-date.
- Data quality rules are fully enforced.